

Smart Mall Parking Management System

Project Type: 48-Hour Hackathon Prototype [cite: 1]

Target Architecture: Full-Stack Web App [cite: 1]

Document Version: 1.0.0

Status: Approved Baseline

Role Scope: Admin, Customer, Security [cite: 1]

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1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) specifies the functional, non-functional, and system architectural requirements for the **Smart Mall Parking Management System** [cite: 1]. The document serves as the primary technical specification for developers, quality assurance testers, and project evaluators during a 48-hour hackathon development cycle [cite: 1].

1.2 Scope

The system is a centralized full-stack web application aimed at modernizing parking operations within shopping malls [cite: 1]. It provides end-to-end management across three primary actor tiers: **Customers** (advance slot reservation, vehicle management, digital QR parking passes) [cite: 1], **Security Staff** (real-time QR verification, gate entry/exit logging) [cite: 1], and **System Administrators** (floor/slot CRUD management, operational analytics) [cite: 1].

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification following IEEE Std 830 principles.
QR Code	Quick Response Code — a 2D matrix barcode utilized for instant pass validation [cite: 1].
CRUD	Create, Read, Update, Delete database operations [cite: 1].
RBAC	Role-Based Access Control enforcing permission boundaries per user tier [cite: 1].
Active Pass	A parking reservation currently in 'Reserved' or 'Occupied' state [cite: 1].

2. OVERALL DESCRIPTION

2.1 Product Perspective

The Smart Mall Parking Management System operates as an autonomous, web-accessible platform replacing traditional manual paper ticketing [cite: 1]. The architecture decouples front-end user interaction from back-end logic and database state persistence, allowing high-speed scanning at physical entry and exit gates via tablet or smartphone browsers [cite: 1].

2.2 Product Functions Overview

The system provides core workflow capabilities summarized in the functional breakdown below:

- Authentication & Authorization:** Multi-role login with secured session tokens [cite: 1].
- Vehicle Registry:** Multi-vehicle profile management under single customer accounts [cite: 1].
- Space Layout Management:** Multi-floor (P1, P2, P3) and slot configuration [cite: 1].
- Reservation & Pass Generation:** Real-time slot locking and dynamic QR payload generation [cite: 1].

- **Gate Operations (Scan Engine):** Instant QR reading for entry validation and automated exit duration/fee calculation [cite: 1].
- **Operational Intelligence:** Administrative visualization of occupancy ratios, customer metrics, and usage history [cite: 1].

2.3 User Classes and Characteristics

User Class	Description & Key Responsibilities	Technical Skill Level & Access Level
Customer	Mall visitors reserving slots, generating QR passes, and viewing history [cite: 1].	Basic consumer web interaction. Responsive mobile/desktop UI access [cite: 1].
Security Staff	Entrance and exit gate personnel verifying digital passes [cite: 1].	Basic technical skills. Access restricted to scanning & gate check-in/out modules [cite: 1].
Administrator	Mall management monitoring operations, slots, floors, and analytical reports [cite: 1].	Moderate to high. Full access to management CRUD operations and system reports [cite: 1].

2.4 Operating Environment

The application target environment is optimized for rapid deployment within a 2-day hackathon setting [cite: 1]:

- **Client Side:** Cross-platform modern web browsers (Google Chrome v100+, Safari v15+, Firefox v100+) on desktop, tablet, and mobile browsers [cite: 1].
- **Server Side:** Node.js / Python Runtime with web application framework (e.g., Express or Django/FastAPI) [cite: 1].
- **Database:** Relational Database Engine (e.g., PostgreSQL / MySQL / SQLite) enforcing ACID compliance [cite: 1].

3. SYSTEM ARCHITECTURE & HIGH-LEVEL WORKFLOW

3.1 End-to-End System Workflow

System State Transition:
Customer Registration/Login → Vehicle Addition → View Available Slots → Reserve Slot (AVAILABLE → RESERVED) → Generate Unique QR → Gate Scan at Entry (RESERVED → OCCUPIED) → Shopping → Gate Scan at Exit (OCCUPIED → AVAILABLE) → Compute Duration & Archival [cite: 1].

3.2 Entity Relationship Model (Data Overview)

Entity	Primary Attributes	Relationships
User	user_id , name, email, password_hash, role (Admin/Security/Customer), phone [cite: 1]	1:N with Vehicles, 1:N with Bookings [cite: 1]
Vehicle	vehicle_id , customer_id, vehicle_number, vehicle_type, owner_name, contact_no [cite: 1]	N:1 with User, 1:N with Bookings [cite: 1]
Floor	floor_id , floor_code (P1, P2, P3), total_slots, status [cite: 1]	1:N with Slots [cite: 1]
Slot	slot_id , floor_id, slot_number, status (Available/Reserved/Occupied) [cite: 1]	N:1 with Floor, 1:N with Bookings [cite: 1]
Booking	booking_id , customer_id, vehicle_id, slot_id, qr_code_hash, status, entry_time, exit_time, duration [cite: 1]	N:1 with User, N:1 with Vehicle, N:1 with Slot [cite: 1]

4. FUNCTIONAL REQUIREMENTS (MODULE SPECIFICATIONS)

Module 1: User Authentication & Role Management

- **FR-1.1:** System shall allow users to register with role assignment (Customer by default) [cite: 1].
- **FR-1.2:** System shall validate user credentials (email/password) and issue role-aware tokens [cite: 1].
- **FR-1.3:** System shall redirect authenticated users to their corresponding dashboard (Admin, Security, Customer) [cite: 1].

Module 2: Vehicle Management

- **FR-2.1:** Customers shall be able to register multiple vehicles providing vehicle number, type, owner name, and contact number [cite: 1].
- **FR-2.2:** System shall support full CRUD operations (Create, Read, Update, Delete) on customer vehicle records [cite: 1].

Module 3 & 4: Floor & Slot Management

- **FR-3.1:** Administrators shall create, update, and remove floor layouts (e.g., P1, P2, P3) [cite: 1].
- **FR-4.1:** System shall manage individual slots per floor, enforcing state transitions between **AVAILABLE**, **RESERVED**, and **OCCUPIED** [cite: 1].

Module 5 & 6: Booking & QR Code Pass Generation

- **FR-5.1:** Customers shall view real-time available slots filterable by floor and reserve a slot in advance [cite: 1].
- **FR-5.2:** Upon booking confirmation, the slot status shall automatically transition from **AVAILABLE** to **RESERVED** [cite: 1].
- **FR-6.1:** System shall instantly generate a cryptographic QR code encoding unique reservation data [cite: 1].

Module 7 & 8: Gate Operations (Vehicle Entry & Exit)

- **FR-7.1:** Security staff shall scan the QR pass at entrance. Upon validity check, system records **entry_time** and shifts slot state from **RESERVED** to **OCCUPIED** [cite: 1].
- **FR-8.1:** Security staff shall scan QR pass at exit. System records **exit_time**, calculates total elapsed duration, and shifts slot state from **OCCUPIED** to **AVAILABLE** [cite: 1].

Module 9, 10, 11, 12: Dashboards & Reporting

- **FR-9.1:** Customers can view complete parking history (booking date, slot ID, entry/exit timestamps, total duration) [cite: 1].
- **FR-10.1:** Customer dashboard displays active bookings, registered vehicles, active QR codes, and history [cite: 1].
- **FR-11.1:** Security dashboard provides live QR scanner interface, check-in/out verification logs, and active vehicle list [cite: 1].
- **FR-12.1:** Admin dashboard displays aggregate capacity stats (Total, Available, Reserved, Occupied), customer growth, and graphical report charts [cite: 1].

5. NON-FUNCTIONAL REQUIREMENTS

5.1 Performance Requirements

- **NFR-1 (QR Processing Latency):** QR Code scanning and gate entry validation response time shall be under 1.5 seconds.
- **NFR-2 (Concurrency):** The backend shall support concurrent booking requests without double-booking a single slot.

5.2 Security Requirements

- **NFR-3 (Role Access Control):** Endpoints must be protected by Role-Based Access Control (RBAC) middleware [cite: 1].
- **NFR-4 (Password Hashing):** All user credentials must be stored securely using standard hashing functions (e.g., bcrypt/Argon2).

5.3 Usability & Hackathon Feasibility

- **NFR-5 (Responsive UI):** Security scanning interface must be optimized for mobile touchscreens and camera input [cite: 1].
- **NFR-6 (Rapid Deployment):** The system must be executable within a single container or local setup for live hackathon demonstration [cite: 1].